

10 Boolean logic

Candidates should be able to:

- 1 Identify and use the standard symbols for logic gates
- 2 Define and understand the functions of logic gates
- 3 (a) Use logic gates to create given logic circuits from a:
 - (i) problem statement
 - (ii) logic expression
 - (iii) truth table
- (b) Complete a truth table from a:
 - (i) problem statement
 - (ii) logic expression
 - (iii) logic circuit
- (c) Write a logic expression from a:
 - (i) problem statement
 - (ii) logic circuit
 - (iii) truth table

Notes and guidance

- See section 4 for logic gate symbols
- Including:
 - NOT
 - AND
 - OR
 - NAND
 - NOR
 - XOR (EOR)
 - the binary output produced from all the possible binary inputs
- NOT is a single input gate
- All other gates are limited to two inputs
- Circuits must be drawn for the statement given, without simplification
- Logic circuits will be limited to a maximum of three inputs and one output
- An example truth table with three inputs, for completion:

A	B	C	Output
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	